

LAB WORKSHEET

Lab 1 — Server Form Factors, Racking, and Power Planning

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Objective: 1.1 — Install physical hardware: form factors, racking, cooling, power planning.

Work through the lab on the Killercoda Ubuntu playground and record your results below.

Task 1 — System identity

Run the dmidecode system/chassis commands.

Record manufacturer, product name, serial and chassis type reported.

Task 2 — Component inventory

Run lscpu, lsmem, lspci and lshw -short.

List CPU model, socket/core count, total RAM and one expansion card.

Task 3 — HCL lookup

Get the product string and check it against a vendor HCL.

Paste the product name and note whether CPU / RAID / NIC are HCL-listed.

Task 4 — Rack elevation

Draft a 42U layout applying the balancing / cooling / power rules.

Sketch U positions for switches, compute, JBOD, UPS and PDUs.

Task 5 — Power budget

Sum nameplate watts for your planned rack and compare to a 30 A @ 208 V circuit.

Record total watts, % of circuit, and whether you must split PDUs.

Completion checklist

- Captured system make/model/serial and chassis type with dmidecode.
- Inventoried CPU, memory, bus types and expansion cards.
- Completed an HCL lookup for the target OS.
- Produced a balanced, cooled 42U rack elevation.
- Verified the power budget stays under 80% of the circuit.